



Accelerated diabetic wound healing using a chitosan-based nanomembrane incorporating nanovesicles from *Aloe barbadensis*, *Azadirachta indica*, and *Zingiber officinale*

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ABSTRACT

Diabetic wounds pose a substantial clinical challenge due to delayed healing, persistent inflammation, and susceptibility to infections. This study investigates the therapeutic efficacy of a chitosan-polyvinyl alcohol nanomembrane (OXY-NM^{Aloe}) plant-derived extracellular vesicles enriched with extracellular vesicles derived from *Aloe barbadensis*, *Azadirachta indica*, and *Zingiber officinale*. Chitosan, a natural biological macromolecule, forms the nanomembrane's matrix, contributing to its flexibility, porosity, and structural integrity, essential for maintaining optimal wound hydration and supporting tissue regeneration. In in vivo studies on streptozotocin-induced diabetic rats, OXY-NM^{Aloe} significantly accelerated wound closure by approximately 23 % compared to just 7 % in the control-treated group, even after one day. This effect was achieved by modulating pro-inflammatory cytokines, activating collagen synthesis, and restoring mitochondrial function. The membrane also inhibited matrix metalloproteinase overexpression, reducing excessive extracellular matrix degradation by ~40 % and promoting tissue regeneration. Furthermore, OXY-NM^{Aloe} demonstrated potent antimicrobial activity against *Staphylococcus aureus* and *Pseudomonas aeruginosa*, decreasing microbial colonization and fostering a favorable healing environment. By integrating the structural properties of chitosan with the bioactivity of plant-derived extracellular vesicles, the nanomembrane offers a multifunctional therapeutic platform for accelerating tissue repair and addressing key challenges in diabetic wound management.

1. Introduction

Foot ulcers caused by diabetes are the most prevalent and severe complications of diabetes, often resistant to treatment through diet, exercise, or insulin therapy [1]. Each year, around 18.6 million people worldwide suffer from diabetic foot ulcers, suggesting the widespread prevalence of this condition [2]. Approximately 15–25 % of individuals with diabetes will develop a foot ulcer at some point in their lives,

highlighting the significant risk posed by this complication [3]. Furthermore, about 20 % of infected diabetic foot ulcers lead to lower limb amputation, highlighting the severe consequences of untreated or poorly managed ulcers, while the cost of managing DFUs in the United States exceeds \$13 billion annually, imposing a significant economic burden on the healthcare system [1]. The combination of high amputation rates and substantial financial strain emphasizes the urgent need for effective prevention and management strategies to address diabetic

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foot ulcers comprehensively.

The complex pathology of diabetes-related foot ulcers is driven by persistent hyperglycemia and its associated complications, leading to compromised skin barriers, increased susceptibility to infections, elevated oxidative stress (OS), and disruption of extracellular matrix (ECM) integrity [4]. These factors, combined with impaired cellular adhesion and microvascular dysfunction, contribute to delayed wound healing and an elevated risk of severe complications [5]. Gaining a thorough understanding of the underlying mechanisms and pathophysiology of diabetic foot ulcers is essential for developing management and effective treatment strategies, particularly those incorporating herbal remedies. Exploring and combining the potential of medicinal plants with proven wound-healing and antioxidant properties can lead to innovative, natural therapies. Such approaches not only address the underlying complications of diabetic ulcers but also offer sustainable and biocompatible solutions for long-term management. Previous studies have explored the role of plant-derived exosome-like nanovesicles in chronic wound healing, highlighting their therapeutic potential [6]. Additionally, research on vesicular nanocarriers for phytocompound delivery has demonstrated their effectiveness in wound care [7]. A number of advanced wound healing approaches have been explored in the past, such as nanofiber-based approaches, hydrogels, foam dressings, alginate dressings, and hydrocolloid dressings [8,9]; however, the potential of plant-derived products and compounds in wound healing is often overlooked, despite their historical significance in therapeutics. There are many gold-standard treatments that contain plant-based compounds, and it may be worthwhile to investigate them further, particularly through nanoformulations [10]. We have recently explored a therapeutic nano-formulation using nano-vesicles derived from aloe vera (*Aloe barbadensis*), ginger (*Zingiber officinale*), and neem fruit (*Azadirachta indica*), OXY-Exo^{Aloe}, specifically designed for treating diabetic wound injuries [11]. Aloe vera is well known for its hydrating, anti-inflammatory, and collagen-boosting properties, which can accelerate wound healing [12]. Neem shows strong antimicrobial, antioxidant, and immunomodulatory effects, which help prevent infections and support tissue regeneration [13]. Ginger provides anti-inflammatory, antioxidant, and circulatory-boosting properties, further aiding the healing process [14]. These nanovesicles collectively enhance the therapeutic potential of OXY-NM^{Aloe} by delivering a synergistic blend of bioactive compounds, contributing to its antimicrobial, anti-inflammatory, and wound-healing properties. This formulation is encapsulated in a chitosan-polyvinyl alcohol (PVA)-based nano-membrane, which is a biological macromolecule known for its biocompatibility, biodegradability, and structural integrity. Chitosan contributes significantly to the membrane's mechanical properties, providing flexibility, porosity, and structural integrity to maintain wound hydration and support tissue regeneration, while PVA enhances flexibility and the membrane's ability to retain moisture at the wound site. This combination optimizes membrane performance, enhancing its protective and healing properties.

In diabetic patients, wounds often struggle to heal due to poor blood circulation, leading to a hypoxic (low oxygen) environment that hampers the healing process [15]. Oxygen is essential for several critical functions, including collagen synthesis, cell migration, and the activation of growth factors that are required for wound closure [16]. It also helps reduce inflammation and prevent infection by supporting immune cell function. Chronic hyperglycemia and poor microvascular circulation in diabetic wounds further compromise oxygen delivery, making tissue repair more challenging.

In this study, we developed a chitosan-PVA nanoformulation membrane (OXY-NM^{Aloe}) designed to provide oxygenation to the wound, a crucial factor in effective healing, particularly in diabetic foot ulcers where hypoxia impedes recovery. Through this membrane, medicinally rich nanovesicles derived from aloe vera, ginger, and neem fruit are effectively delivered to the wound site. Their paracrine signaling enhances healing by improving cellular metabolism, reducing oxidative

stress, and stimulating tissue regeneration. The membrane acts as a conduit, allowing therapeutic components to reach the wound and create an optimal environment for faster, more efficient healing.

2. Material and methods

2.1. Animal model for diabetic wound

The study protocols and procedures received approval from the Institutional Animal Ethical Committee (IAEC) of Chhatrapati Shahu Ji Maharaj University, Kanpur (IAEC Approval number: IAEC/UIP/2024/041). Streptozotocin (STZ)-induced diabetic male Sprague-Dawley rats (230–250 g, 10–12 weeks old) were employed to study diabetic wound healing. Throughout the experiment, the rats were maintained under controlled environmental conditions, including a temperature range of 24–25 °C, relative humidity of 45–55 %, and a 12-hour light-dark cycle. Standard laboratory pellets and water were provided ad libitum to ensure the animals' well-being during the study period. Animal experiments were conducted in accordance with ARRIVE guidelines.

2.2. Induction of diabetes

Diabetes mellitus was induced in the experimental rats by administering two intraperitoneal injections of STZ at a dose of 45 mg/kg on days 1 and 3. The STZ solution was freshly prepared in a cold 0.1 M sodium citrate buffer (pH 4.5) immediately before administration. Prior to the injections, the animals were fasted for approximately 16 h, with water provided freely. Blood glucose levels were measured using tail-vein blood samples and a glucometer. Rats with blood glucose levels exceeding 300 mg/dL were considered diabetic.

2.3. Generation of diabetic wound rat model

Diabetic animals were restrained and anesthetized using an intraperitoneal injection of ketamine (85 mg/kg) combined with xylazine (6 mg/kg). Before creating the wounds, the dorsal fur was carefully shaved and the skin disinfected with 70 % ethanol. An open excision wound (5 mm in diameter and 2 mm in depth) was created using a biopsy punch and surgical scissors. Skin samples from the wound site were collected immediately, placed in chilled tubes, and stored at –80 °C for subsequent biochemical and molecular analyses.

2.4. OXY-NM^{Aloe} preparation and application

The films were prepared using the solvent casting method with chitosan and polyvinyl alcohol (PVA). Chitosan was dissolved in 1 % aqueous acetic acid and stirred for 2 h to obtain a homogeneous solution. A 5 % PVA solution was prepared by stirring in distilled water for 2 h. PVA was then added to the chitosan solution (30/70 chitosan/PVA ratio) and mixed for 1 h [17]. OXY-Exo^{Aloe}, mixed in aloe vera extract (200 µL), was added to chitosan/PVA mixtures and sonicated to remove air bubbles. For control membranes (CON-M) preparation, the same volume of aloe vera extract was added. The two solutions were separately poured into a 10 cm petri dish and dried at 40 °C for 5.5 h, with hourly weight measurements until a constant weight was achieved. After drying, the films were peeled off and stored for further testing. The films' mechanical properties, including tensile strength, elongation at break, swelling index were then evaluated as described earlier [17]. Detailed protocol is provided in Supplementary Methodology S-2.1.

2.5. Scanning electron microscope analysis

Membrane samples (1–2 cm) were rinsed with distilled water and air-dried. They were then immersed in 2.5 % glutaraldehyde for 1 h at room temperature for fixation. After fixation, the samples were sequentially dehydrated through ethanol solutions, followed by drying

with a critical point dryer to prevent structural collapse. Once dried, the samples were sputter-coated with gold to ensure conductivity and subsequently analyzed using scanning electron microscopy (SEM) (JEOL Ltd., MA, USA).

2.6. Biochemical and molecular assays

2.6.1. Cytokine assay

After three days of OXY-NM^{Aloe} treatment, cotton swabs were used to collect samples from the wounds of both treated and untreated diabetic rats for cytokine analysis. Tumor necrosis factor- α (TNF- α) levels were measured using an enzyme-linked immunosorbent assay (ELISA) (BIORAD, CA, USA).

2.6.2. RT-PCR assay

RT-PCR was conducted to assess the expression levels of the inflammatory markers, mitochondrial respiratory complex genes, and ECM-related genes. In brief, total RNA was extracted from wound tissue using TRIzol® reagent (Invitrogen, USA) following the manufacturer's guidelines. Complementary DNA (cDNA) synthesis was performed with the iScript cDNA kit (BIORAD), and RT-PCR was conducted using the GoTaq® green master mix (Promega, USA). The resulting PCR products were visualized on an ethidium bromide-stained agarose, and images were captured using a gel documentation system (BIORAD). Band intensities of the PCR products were quantified and normalized to β -actin expression for comparison. List of primers is provided in Supplementary Table 1, S-1.

2.6.3. Biochemical assays

Biochemical assays were carried out as previously described [11,18]. In brief, skin samples from rat wounds were homogenized and used to evaluate lipid peroxidation and glutathione levels.

2.6.4. Nitrite levels

Total nitrite content in wound samples was estimated as described previously using Griess reagent (1 % N-(1-Naphthyl) ethylenediamine dihydrochloride) [19]. Equal volumes of Griess reagent and wound tissue samples were treated with 1 mM NADH, nitrate reductase (1.0 U/ml), and 1 M zinc acetate. Samples were then incubated at 37 °C for 10 min, followed by absorbance measurement at 542 nm using an ELISA plate reader (BIORAD).

2.7. Antimicrobial activity

The antimicrobial activity was assessed using the agar diffusion method [20] against four pathogenic bacteria: *Staphylococcus aureus*, *Escherichia coli*, *Bacillus subtilis*, and *Pseudomonas aeruginosa*. The bacterial strains were obtained from the Department of Microbiology at Chhatrapati Shahu Ji Maharaj University, Kanpur, and were cultured in nutrient broth (Himedia, Mumbai) at 30 °C. Overnight cultures were evenly spread on Muller Hinton agar (MH) plates. CON-M and OXY-NM^{Aloe} membranes were cut into squares and placed on the surface of the agar. The plates were sealed with Parafilm and incubated at 37 °C for 24 h. After incubation, the zone of inhibition around the membrane was measured to evaluate antibacterial activity, with the clear zone indicating the extent of bacterial inhibition.

2.8. Statistics analysis

The data obtained from this experiment were analyzed statistically using GraphPad Prism 9.4.1. The results are presented as the mean $\pm$ standard error of the mean (SEM). For comparisons between two groups, the Student's *t*-test was employed. Welch's unpaired *t*-test was used for statistical comparisons between two groups, where the dataset showed unequal variances and involved small sample sizes. A *p*-value <0.05 was considered statistically significant.

Some methodological details are provided in the Supplementary Information.

3. Results

3.1. Generation and physicochemical characterization of OXY-NM^{Aloe} nanomembranes

Fig. 1 presents pictorial representations of two different chitosan/PVA membranes: one incorporated with *aloe vera* extract and OXY-Exo^{Aloe} nanovesicles (OXY-NM^{Aloe}) and the other without nanovesicles (CON-M).

These OXY-NM^{Aloe} membranes are transparent, homogeneous, and inflexible, exhibiting excellent consistency with a uniform thickness (Fig. 2A, B, C). The two-hydrogel membrane exhibited a skin-friendly pH (5.5–5.9) and an excellent folding endurance (>300). Both membranes also exhibited high moisture retention, a strong swelling index, and favorable physicochemical properties, suggesting that the membrane is ideal for prolonged use without losing structural integrity, while maintaining an optimum wound environment for wound healing applications (S-3, Supplementary Figs. 1, 2). SEM analysis of the OXY-NM^{Aloe} revealed a smooth, uniform surface with no visible cracks or irregularities. The OXY-NM^{Aloe} membrane exhibited well-defined, ovoid, and interconnected pores of varying sizes and shapes (Fig. 2D). The porous structure and surface morphology of the gold-coated membrane confirmed the successful integration and uniform distribution of OXY-Exo^{Aloe}, highlighting its structural integrity and potential for targeted drug delivery and wound healing (Fig. 2E).

3.2. Accelerated wound healing in diabetic rats using OXY-NM^{Aloe} nanomembranes

OXY-NM^{Aloe}, along with CON-M, was applied to different wounds 1 h after their creation in STZ-induced diabetic rats. Physical analysis of the wound sites was conducted to evaluate the treatment's effectiveness. Throughout the treatment period, no significant changes in body weight were observed, and the rats displayed no adverse effects. The results showed that the OXY-NM^{Aloe} accelerated wound healing in STZ-induced diabetic rats compared to the CON-M-treated group, with noticeable improvements as early as 24 h (Fig. 3A, B). Treated wounds over time exhibited a lower number of red blood cells (RBCs), smaller wound size, and increased collagen deposition and epithelialization, indicating enhanced healing (Fig. 3B, C, D). Compared to CON-M, the wound closure rate was significantly augmented in the OXY-NM^{Aloe}-treated group, as reflected in the heat plot (Fig. 3E), and wound closure analysis (Fig. 3F, G). The wound closure rate was significantly augmented in the OXY-NM^{Aloe}-treated group compared to CON-M at all assessed time points - day 1, day 3, day 7, and day 14 - with significantly high statistical significance (*p* < 0.0001), as quantified in the wound closure analysis (Fig. 3F). These results demonstrate that the OXY-NM^{Aloe} facilitated a faster healing process compared to the control membrane treated group, with a noticeable reduction in wound size over time.

3.3. OXY-NM^{Aloe} treatment lowers BWAT scores

The Bates–Jensen Wound Assessment Tool (BWAT) is commonly used to evaluate wound healing, with a score of 13 or below signifying wound regeneration and scores above 13 indicating wound deterioration. Compared to CON-M-STZ rats, OXY-NM^{Aloe}-STZ rats demonstrated significantly lower BWAT scores within seven days, reflecting an accelerated wound healing process in OXY-NM^{Aloe} group (Fig. 4, Supplementary Fig. 3).

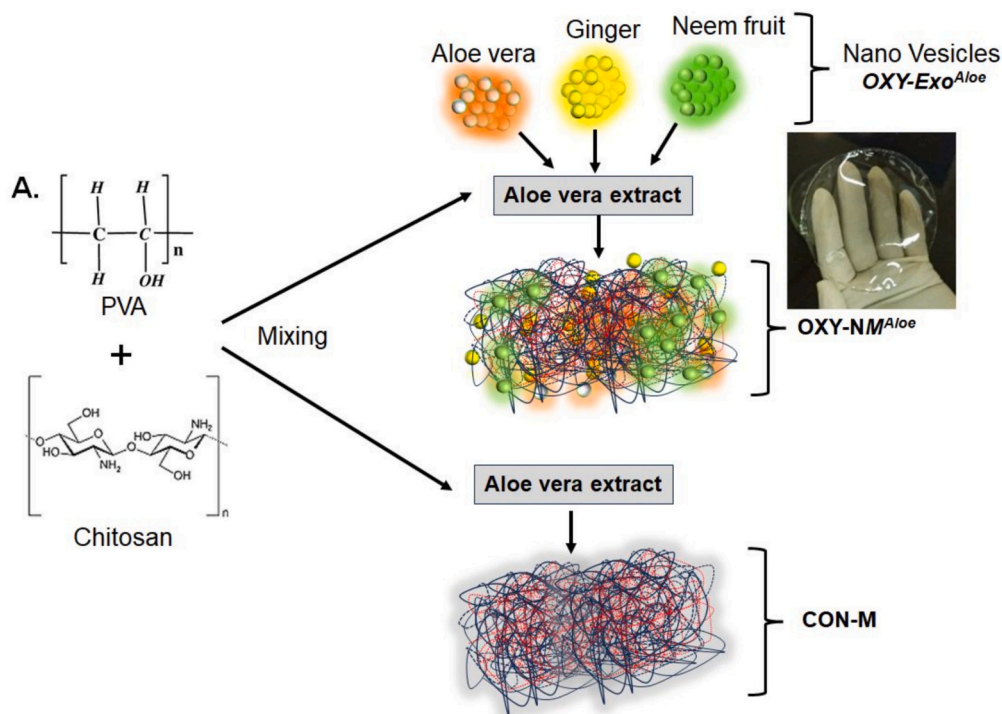


Fig. 1. (A) Schematic representation of CON-M, prepared with *aloe vera* extract, and OXY-NM^{Aloe}, incorporating aloe vera extract along with nanovesicles derived from *Aloe barbadensis*, *Azadirachta indica*, and *Zingiber officinale*.

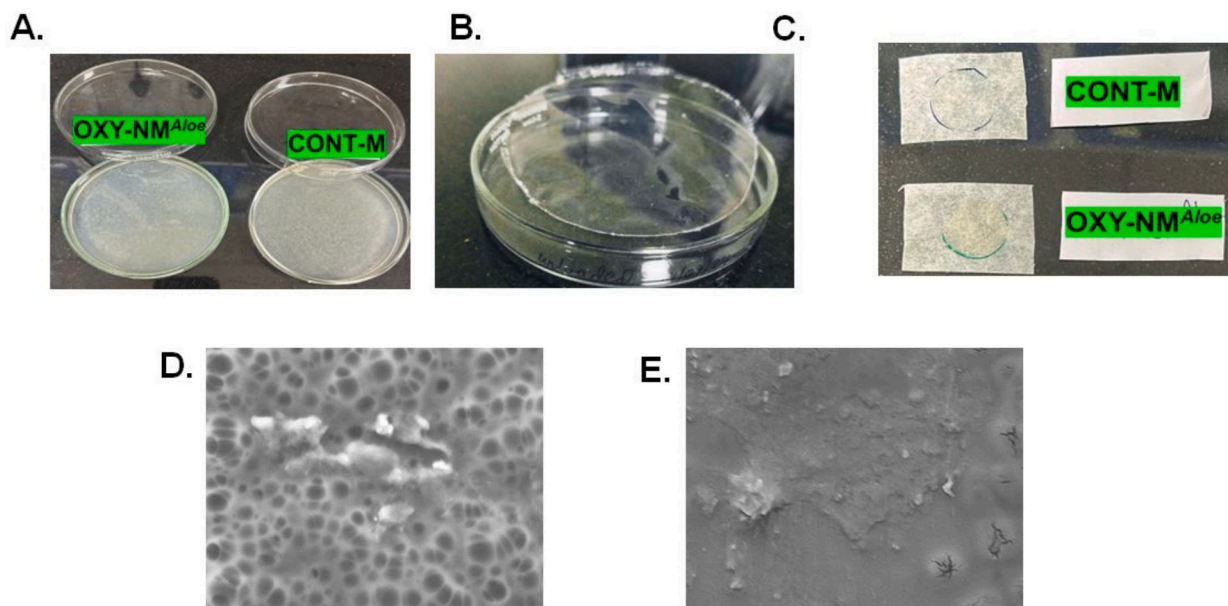


Fig. 2. (A, B) Representative images showing CON-M and OXY-NM^{Aloe} hydrogel preparation. (C) Representative image showing the covering pad. (D, E) Scanning Electron Microscopy images illustrating the porous membrane structure embedded with nanovesicles. A thin layer of gold coating was applied to enhance surface conductivity, enabling clear visualization of membrane porosity and nanovesicle distribution.

3.4. Normalization of protease, inflammatory, and mitochondrial stress by OXY-NM^{Aloe} treatment

Efficient wound healing and regeneration depend on the precise regulation of gelatinase activity. In comparison to CON-M-treated diabetic rats, OXY-NM^{Aloe} treatment significantly reduced the transcript expressions of MMP-2 ($p = 0.0257$) and MMP-9 ($p = 0.0127$) (Fig. 5A, B), thereby promoting improved wound healing by normalizing

protease stress. STZ-induced diabetic rats treated with CON-M exhibited elevated IL-6 transcript levels, which were significantly reduced following OXY-NM^{Aloe} treatment ($p = 0.0045$), indicating attenuation of inflammatory stress. Contrary to this, OXY-NM^{Aloe} treatment led to a significant increase in Transforming Growth Factor-beta (TGF- β) expression ($p = 0.0027$), supporting enhanced angiogenesis and wound healing (Fig. 5A, B). Furthermore, mitochondrial respiratory complex transcript levels were assessed to evaluate the effect of the nano-therapy

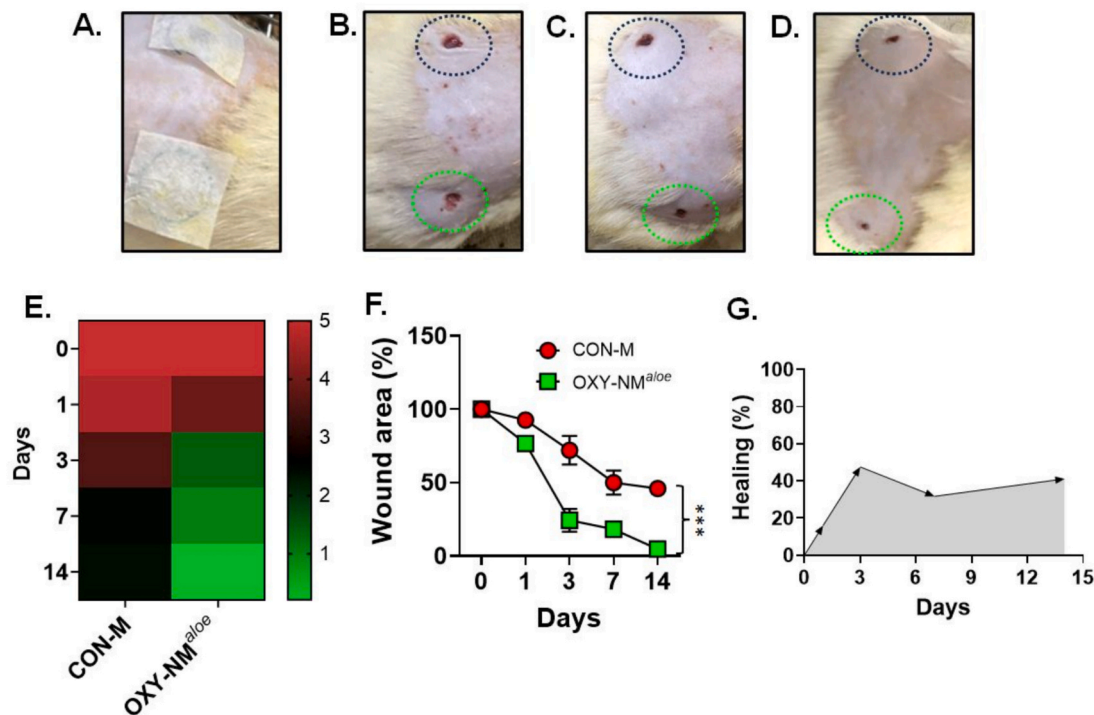


Fig. 3. (A) Image showing the dorsal side of diabetic rats with two wounds covered with CON-M and OXY-NM^{Aloe}. (B-D) Time-course images displaying wound healing progression in diabetic rat groups at 1 day (B), 3 days (C), and 7 days (D). The black dotted circle marks the wound covered with CON-M, while the green dotted circle represents the wound treated with OXY-NM^{Aloe}. (E) Heatmap illustrating healing rates in diabetic rat groups treated with CON-M and OXY-NM^{Aloe}. Each block is the median of group data collected on different days. (F) Line plot depicting the wound closure rate in CON-M and OXY-NM^{Aloe} treated groups. *** $p < 0.0001$ vs-CON-M-treated rat group in all time points. (G) Plot showing the percentage improvement in wound healing at different time points in the OXY-NM^{Aloe}-treated group compared to the CON-M-treated group.

	CON-M				OXY-NM ^{Aloe}			
Days	1	3	7	15	1	3	7	15
Size (length X width)	1	1	1	1	1	1	1	1
Depth	3	2	2	2	3	2	1	1
Edges	3	3	2	2	2	2	1	1
Undermining	2	2	2	2	2	1	1	1
Nacrotic tissue type	4	3	3	1	3	2	1	1
Nacrotic tissue amount	4	3	2	1	3	1	1	1
Exudate type	3	3	2	1	2	2	1	1
Exudate amount	2	2	2	2	1	1	1	1
Skin color surrounding wound	3	2	1	1	2	1	1	1
Peripheral tissue edema	3	3	1	1	2	2	1	1
Peripheral tissue induration	1	1	2	2	1	1	1	1
Granulation tissue	4	3	2	2	3	2	1	1
Epithelialization	4	3	2	2	3	1	1	1
Total Score	37	31	24	20	28	19	13	13

Fig. 4. Heatmap illustrating the effectiveness of OXY-NM^{Aloe} and CON-M treatments in wound evaluation across diabetic rat groups based on various Bates-Jensen Wound Assessment criteria. The color gradient represents the severity of each parameter, with darker shades indicating higher intensity.

on mitochondrial recovery at the diabetic wound site. In OXY-NM^{Aloe}-treated STZ rats, transcript levels of mitochondrial NADH-ubiquinone oxidoreductase chain 1 (ND1) were significantly upregulated ($p = 0.0114$), indicating improved mitochondrial integrity. Although NDUFB8 expression appeared elevated, the difference was not statistically significant ($p = 0.0536$), suggesting a positive trend toward enhanced mitochondrial function (Fig. 5A, B). In addition, TNF- α transcript levels were markedly reduced following OXY-NM^{Aloe} treatment, showing a significant decrease ($p = 0.0275$) compared to the CON-M

group. This reduction reflects a strong anti-inflammatory response induced by the nanoformulation, with expression levels dropping by over 40 % in treated wounds (Fig. 5C).

3.5. Mitigation of oxido-nitrosative stress by OXY-NM^{Aloe} treatment

Next, we assessed oxido-nitrosative stress, a critical factor impeding diabetic wound healing. Malondialdehyde (MDA) levels, an indicator of lipid peroxidation and oxidative damage, were significantly elevated in

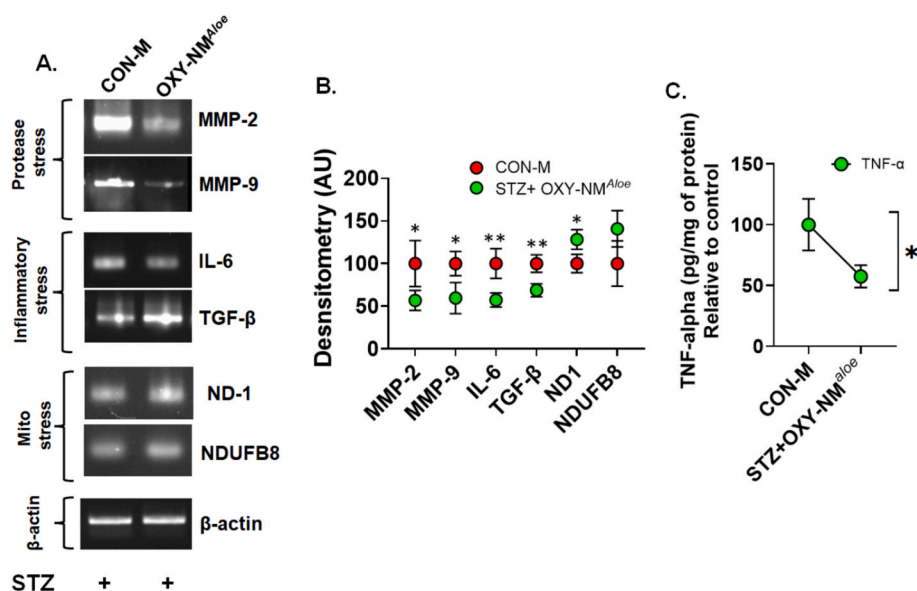


Fig. 5. (A) Agarose gel images showing transcript levels of genes associated with inflammation, oxidative stress, matrix remodeling, and mitochondrial function in diabetic rats treated with CON-M and OXY-NM^{Aloe}. (B) Densitometric analysis of gene expression levels for MMP-2, MMP-9, IL-6, TGF-β, ND1, and NDUFB8. * $p < 0.05$, ** $p < 0.005$ vs-CON-M treated rat group. (C) Error and bar plot representing TNF-α levels in the CON-M and OXY-NM^{Aloe} treated diabetic rat wounds, analyzed by ELISA. * $p < 0.05$ vs-CON-M-treated rat group.

the wound tissues of CON-M-treated diabetic rats. However, OXY-NM^{Aloe} treatment led to a reduction of approximately 37 % in MDA levels ($p = 0.0362$), indicating alleviation of oxidative stress (Fig. 6A). In contrast, glutathione (GSH) levels, severely reduced in the CON-M group, showed a substantial increase upon OXY-NM^{Aloe} treatment ($p = 0.0189$), suggesting improved redox balance (Fig. 6A). Similarly, nitrite levels, elevated in the diabetic wounds, were significantly reduced by ~42.5 % following OXY-NM^{Aloe} treatment ($p = 0.0275$), reflecting effective mitigation of nitrosative stress (Fig. 6B). Together, these results demonstrate that the OXY-NM^{Aloe} nanoformulation successfully counteracts oxido-nitrosative stress, promoting a more favorable environment for wound repair.

3.6. Antimicrobial activity of OXY-NM^{Aloe} nanomembrane against wound pathogens

The antimicrobial activity of the OXY-NM^{Aloe} nanomembrane was evaluated against *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Escherichia coli*, and *Bacillus subtilis*. Compared to CON-M group, the OXY-NM^{Aloe} nanomembrane significantly enhanced bacterial inhibition. A notable increase in the zone of inhibition was observed against *S. aureus* ($p = 0.0012$) and *P. aeruginosa* ($p = 0.0081$) (Fig. 7A, B). It showed significant inhibitory effects against *P. aeruginosa* and *S. aureus*, two common pathogens associated with wound infections [21], and the observed inhibition highlights the nanomembrane's therapeutic potential. However, no significant inhibition was observed against *E. coli* and *B. subtilis*, suggesting organism-specific antimicrobial efficacy (data not

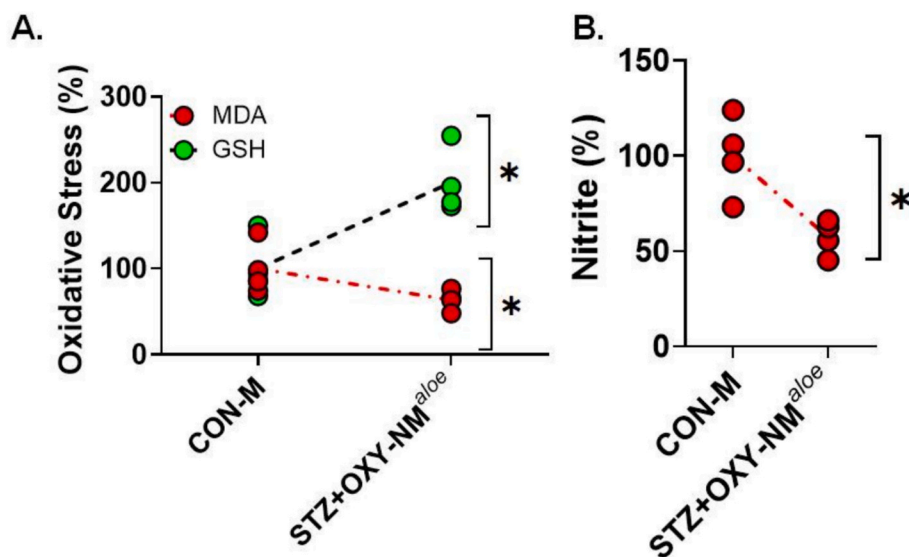


Fig. 6. (A) Plot showing malondialdehyde (MDA) and glutathione (GSH) levels in different rat groups. Red dots represent individual MDA values, while green dots indicate individual GSH values. * $p < 0.05$ vs-CON-M-treated rat group. (B) Plot showing nitrite levels in diabetic wounds treated with CON-M and OXY-NM^{Aloe}. Each rat dot represents an individual rat's value. * $p < 0.05$ vs-CON-M-treated rat group.

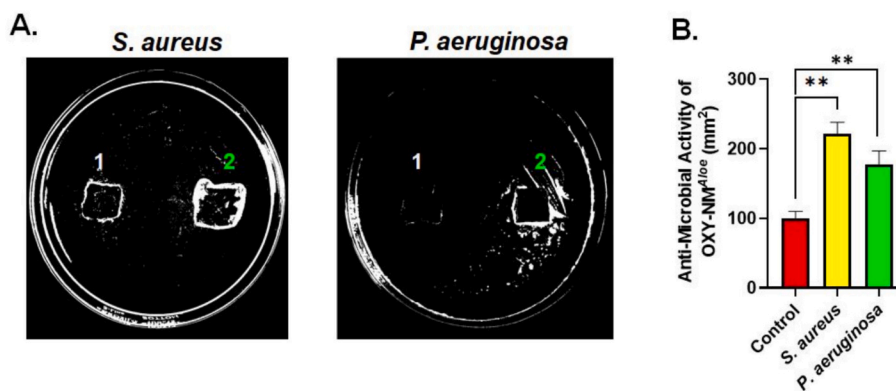


Fig. 7. (A) Representative images showing zones of inhibition against *Staphylococcus aureus* and *Pseudomonas aeruginosa*. Zones of inhibition are shown, highlighting the enhanced antimicrobial effect of OXY-NM^{Aloe}. Here, 1, 2 represent CON-M and OXY-NM^{Aloe}, respectively. (B) Bar graph showing quantitative analysis of bacterial inhibition zones. Data are presented as mean $\pm$ SEM, $n = 3$. ** $p < 0.01$ vs.-CON-M.

shown).

4. Discussion

The present study highlights the efficacy of a novel OXY-Exo^{Aloe} (nanovesicles isolated from *aloe vera*, neem fruit, and ginger) nanomembrane, OXY-NM^{Aloe}, incorporated into a chitosan-PVA-based matrix, in accelerating wound healing in STZ-induced diabetic rats. Diabetic wounds are characterized by prolonged inflammation, delayed cellular proliferation, impaired angiogenesis, and persistent oxidative stress, all of which contribute to chronic non-healing ulcers. Our findings demonstrate that OXY-NM^{Aloe} promotes diabetic wound healing due to its diverse properties, including antimicrobial activity, highlighting its significant therapeutic potential. It reduces oxidative stress and inhibits pro-inflammatory cytokines, creating an optimal environment for tissue repair. Furthermore, we believe that its porous structure facilitates more oxygen delivery through nano-formulations to hypoxic tissues, supporting mitochondrial function and accelerating the wound healing process.

Earlier, we have reported nanoformulation OXY-Exo^{Aloe}, derived from the plant-derived EVs of *Aloe barbadensis*, *Azadirachta indica*, and *Zingiber officinale*, exhibited an average size of ~ 230 nm as determined by NTA analysis. The formulation, with a favorable zeta potential, significantly accelerated wound healing in STZ-induced diabetic rats, reducing the healing time significantly [11]. Building upon this foundation, we engineered OXY-NM^{Aloe} to integrate the nanovesicles within a chitosan/PVA matrix, aiming to enhance wound healing therapeutic efficacy (Supplementary Fig. 4). Alongside, we developed a control membrane with *aloe vera* extract containing only *aloe vera* extract (CON-M) for comparison. *Aloe vera*-based hydrogels have demonstrated properties that confirm their potential for wound healing applications [22]. Previously, chitosan/PVA-based membranes have also been developed for skin applications [23,24]. To characterize the nanomembrane properties, SEM has traditionally been used to analyze surface morphology, size distribution, and porosity, providing valuable insights into the structural characteristics and performance of the membrane [25,26]. Our analysis of the OXY-NM^{Aloe} membrane revealed a smooth, uniform surface with well-defined, interconnected pores, confirming the successful integration and uniform distribution of OXY-Exo^{Aloe}. In addition, nanomembrane exhibited desirable physicochemical properties, including uniform thickness, skin-friendly pH, moisture retention, and excellent folding endurance. These characteristics support its suitability for long-term wound applications while maintaining structural integrity. The therapeutic efficacy of OXY-NM^{Aloe} was further demonstrated through accelerated wound closure and improved healing outcomes in STZ-induced diabetic rats, as assessed by the BWAT score, a method effectively utilized in previous studies to provide a detailed

assessment of the wound healing process [11,27]. This suggests that the membrane, incorporating the OXY-Exo^{Aloe} formulation, plays a crucial role in promoting wound healing in diabetic conditions. We believe that medicinal plant-derived nanovesicles deliver macromolecular paracrine factors that regulate nearby cellular responses, promoting tissue regeneration and accelerating wound healing.

Wounds with poor healing outcomes are often characterized by the overexpression of matrix metalloproteinases (MMPs), particularly gelatinases such as MMP-9 and MMP-2. Gelatinases, primarily matrix metalloproteinases (MMP-2 and MMP-9), play a crucial role in wound healing by degrading ECM components such as collagen and gelatin. This breakdown facilitates tissue remodeling, cell migration, and re-epithelialization [28]. Proper regulation of gelatinase activity is vital for effective wound repair and regeneration. While essential for normal healing, excessive gelatinase activity can impair healing and contribute to chronic wounds [29,30]. Like proteases, cytokines also play an important role in wound healing, aiding inflammation and tissue repair [31]. Although inflammation is necessary for initiating the healing process by recruiting immune cells, chronic overexpression of cytokines in diabetic wounds can prolong inflammation, creating an unfavorable microenvironment that impairs healing. Our study demonstrated that as compared to CON-M-treated diabetic rats, OXY-NM^{Aloe} treatment significantly normalized proteases response, reduced pro-inflammatory cytokine levels, indicating its strong anti-inflammatory properties. This reduction contributes to creating a favorable environment for wound healing by minimizing prolonged inflammation in diabetes. TGF- β plays a crucial role in wound healing by regulating inflammation, promoting fibroblast proliferation, enhancing collagen synthesis, and facilitating tissue remodeling for scar formation [32]. Elevated cytokine levels, which exacerbate oxidative stress and impair TGF β -induced wound healing, hinder the formation of new blood vessels critical for oxygen and nutrient delivery to healing tissues, thereby aggravating diabetic wound healing. Interestingly, our findings revealed that OXY-NM^{Aloe} treatment augmented TGF β levels while concurrently reducing IL-6, highlighting its dual role in promoting angiogenesis and accelerating wound healing. Our data revealed that OXY-NM^{Aloe}-treated wounds exhibited enhanced expression of mitochondrial respiratory complexes ND1 and NDUF8, suggesting that restored mitochondrial function could be one of the reasons for faster tissue recovery by promoting improved bioenergetic capacity and efficient cellular respiration.

Next, we examined oxidative stress, a significant barrier to diabetic wound healing, where elevated levels of reactive oxygen species cause damage to cellular components and extracellular matrices [4,33]. The nanomembrane played a crucial role in mitigating oxidative and nitrosative stress by minimizing oxidative damage and maintaining redox balance, thereby creating an optimal environment for tissue repair and regeneration. The increased oxygenation observed with OXY-NM^{Aloe}

treatment correlated with improved epithelialization, collagen deposition, and accelerated healing, promoting faster and more effective wound recovery in diabetic conditions.

The OXY-NM^{Aloe} membrane exhibited significant antimicrobial activity, particularly against *Pseudomonas aeruginosa* and *Staphylococcus aureus*, two common wound pathogens [21], while demonstrating no notable effects against *Escherichia coli* and *Bacillus subtilis*. The selective antimicrobial activity of the nanomembrane could be likely due to differences in bacterial cell wall structure, membrane permeability, and resistance mechanisms [34]. We believe that the nanomembrane is protective in healing diabetic wounds by effectively reducing microbial load at the wound site. By eliminating harmful microbes, the nanomembrane minimizes inflammation, prevents further tissue damage, and creates a sterile environment conducive to cellular proliferation, collagen synthesis, and tissue regeneration. This microbial clearance could enhance overall wound healing outcomes.

Although there are studies that use exosomes derived from bone marrow [35], mesenchymal stem cells [36], epidermal stem cell (ESCs) [37], human amniotic epithelial cells (hAEC) [38] and endothelial cells [39] but there are no studies that have used exosomes derived from natural medicinal plants embedded in membrane, in diabetic wound healing, which makes this study novel. A study by Li et al. [40] elucidates the use of curcumin-loaded macrophage-derived exosomes in diabetic wound healing. They have shown that Exos-cur can inhibit inflammation by activating the Nrf2/ARE pathway while in our study we have shown that naturally derived exosomes accelerate diabetic wound healing by reducing oxidative stress, reducing the expression of MMPs, reducing mitochondrial stress and inflammatory stress. However, Li et al. have not used a membrane system to embed the exosomes as we have used in the present study. Another very interesting study by Liao et al. [41] uses exosome-like nanoparticles derived from ancient medicinal insect *Periplaneta americana* L. for accelerating diabetic wound healing. They demonstrated that miRNA cargoes in derived exosomes participate in the wound healing related signal pathways such as TGF-beta, mTOR, and autophagy. In spite of using cell derived exosomes or artificial exosomes loaded with drugs, the natural plant derived exosomes are advantageous in terms of low toxicity, low immunogenicity, safety, biocompatibility and stability. This adds value to the present study.

4.1. Conclusion

In conclusion, the OXY-Exo^{Aloe} impregnated in a chitosan/PVA-based nanomembrane OXY-NM^{Aloe} effectively addressed multiple barriers to diabetic wound healing through its antimicrobial, antioxidant, anti-inflammatory and oxygenated properties. The chitosan/PVA matrix plays a critical role in ensuring structural integrity, moisture retention, and stabilization of medicinal nano-vesicles, creating a bioactive and sterile environment conducive to healing. By reducing oxidative stress, enhancing oxygenation, and promoting a sterile, bioactive environment, this nanomembrane offers a promising therapeutic approach for managing chronic diabetic wounds. This study contributes to the United Nations Sustainable Development Goals, particularly SDG 3 (Good Health and Well-being) by promoting advanced wound healing solutions, and SDG 9 (Industry, Innovation, and Infrastructure) through the development of nanotherapeutic biomaterials.

4.2. Strength and limitations of the study

The strength of this study lies in the development and comprehensive characterization of the OXY-NM^{Aloe} nanomembrane, which demonstrated multifunctional therapeutic potential. By effectively mitigating oxidative, nitrosative, protease, and inflammatory stress, the nanomembrane facilitated accelerated wound healing in diabetic conditions. The integration of bioactive compounds from medicinal plants enhanced angiogenesis, improved mitochondrial function, and exhibited selective

antimicrobial properties, making it a promising candidate for advanced wound care.

Despite the promising results, this study has certain limitations. The in vivo experiments were conducted on a STZ-induced diabetic rat model, which may not fully replicate the complexity of human diabetic wounds. Additionally, while the nanovesicle-integrated chitosan/PVA nanomembrane demonstrated significant antimicrobial and wound-healing potential, further clinical studies are required to validate its efficacy and biocompatibility in human subjects. The exact mechanistic pathways through which the nanovesicles exert their therapeutic effects remain to be fully elucidated, necessitating deeper molecular investigations. Future studies are warranted to optimize large-scale nanomembrane production, enhance clinical translation, and assess the long-term stability of OXY-NM^{Aloe} to fully realize its potential in wound care and improve outcomes in managing chronic diabetic wounds.

CRediT authorship contribution statement

Mumtaz Bano Miya: Writing – original draft, Investigation. **Ashutosh:** Writing – original draft, Investigation. **Maulishree:** Methodology, Investigation. **Dhananjay Dey:** Validation, Supervision, Investigation. **Vandana Pathak:** Methodology, Investigation. **Ekta Khare:** Writing – original draft, Investigation. **Komal Kalani:** Writing – review & editing, Validation. **Poonam Chaturvedi:** Validation, Data curation. **Vimal Singh:** Writing – review & editing, Writing – original draft. **Pankaj Chaturvedi:** Writing – review & editing, Supervision. **Anuradha Kalani:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2025.143169>.

Data availability

Data will be made available on request.

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